Juan de Dios Alvarez

Data Scientist -- Economic Research and Decision Analytics

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Profile

Data scientist with seven years of industry work (2019 to present) and a research background in physics and computer science. My most recent institutional role was Economic Research Specialist at FIRA, Mexico's agricultural development bank, where I replaced a manual, single-analyst reporting process with SQL-backed indicators, executive reports and dashboards used by decision-makers. Across roles, my work starts by defining the metric, measuring a method against a baseline or control set before trusting it, and writing the result for a non-technical reader. I work AI-first: I use Claude daily for analysis, code and review, inside written workflows that keep every output traceable to its inputs and checked by deterministic tests.

Professional Experience

Data Scientist & ML Engineer | Independent Practice

March 2026 -- present

- Designed and run, alone, the data system of a small retail business in production: a PostgreSQL warehouse (40+ tables, PL/pgSQL functions and triggers), FIFO cost-layer inventory valuation, and bank-to-invoice-to-payment reconciliation with confidence tiers, used daily by store staff.
- Defined each business metric (stock value, on-hand quantity, trailing sales) once in shared code so reports and screens agree, and wrote data coverage and assumptions into every monthly report so the owner reads figures with their limits.
- Evaluated document-extraction methods head to head on the same documents with character and word error rates against ground truth; adopted or rejected each change from measured before/after pass rates (for example, doubtful lines reduced from 175 to 77 on a fixed sample).
- Built a regulatory-review method for anti-money-laundering compliance: recall@k of keyword, embedding and hybrid retrieval against a labeled question bank, a labeled control set with an explicit acceptance criterion, and a stratified random sample to estimate what the filter missed.

Economic Research Specialist (Data Scientist) | FIRA

January 2025 -- January 2026

- Replaced one analyst's manual Excel process with a scheduled, unattended pipeline: 74 extraction scripts across 9+ public sources (INEGI, Banxico, SNIIM, SIAP, USDA) loading PostgreSQL with provenance metadata.
- Wrote 60+ PostgreSQL functions and indexed materialized views (window functions, JSONB, CTEs) that compute economic indicators: month-over-month inflation and weighted contributions to CPI, trade balances, world supply-and-demand balances, and crop yields.
- Delivered automated executive Excel reports and 15+ Apache Superset dashboards to an economic research unit; translated the unit's dashboard requests into an indicator-by-indicator plan with data availability and gaps.
- Built a crop-production estimation pipeline from Sentinel-2 and Planet Labs vegetation-index time series: cloud-cover thresholds compared by records kept, per-plot outlier filtering, PCA variable selection, and Random Forest and Gradient Boosting models selected by 5-fold cross-validation.
- Set the governance conventions the pipeline runs on (raw data preserved, lineage per extraction, GitLab) and supervised an undergraduate student team contributing to it.

Data Scientist | CITEIM

October 2024 -- January 2025

- Scoped a Mexican Sign Language recognition system from a structured literature review and designed an on-device, two-stage approach.

- Built the data foundation: video sourcing through search APIs, weak labeling from subtitles (OCR and speech-to-text), and pose and hand keypoint extraction with MediaPipe.

Data Scientist & Team Lead | Punto Singular

December 2023 -- August 2024

- Led monthly analytics cycles for an EdTech client: define the problem and its metric, analyze, propose, follow implementation, and present results against the agreed metric.
- Led KPI and quality dashboards (R, Python, Tableau, Looker) used directly by business stakeholders.
- Founded and led a Data Science group of undergraduate students, pairing training with real client and nonprofit work, including a water-quality analysis for an environmental initiative.

Data Scientist | BairesDev

September 2023 -- November 2023

- Built a retrieval-augmented generation proof of concept so a legal client's lawyers could query contract portfolios in plain language, with graph modeling (Neo4j) and transformer-based summarization.

Data Scientist | D3Clarity (formerly Viqtor Davis North America)

February 2019 -- March 2023

- First data scientist of the company; defined the analytics service roadmap and moved it to client delivery for US-based clients.
- Classified public school-board meeting minutes with NLP on Azure ML and delivered a Power BI dashboard the client's sales team used directly.
- Built forecasting models to identify potential consumer segments for client sales and marketing strategy; data lakes, automated ingestion and ETL.

Lecturer & Mentor | Tecnologico de Monterrey, Campus Morelia

2013 -- 2014; 2023 -- 2024

- Taught KPI dashboarding (Tableau) at the Business School to students from economics and management, plus mathematics, physics and programming; mentored an undergraduate robotics team.

Education

- **PhD Candidate in Physics** -- University of Michoacan, Mexico (2009 -- 2013). Large-scale data analysis for the HAWC Collaboration; co-author on 14 publications.
- **Master's in Computer Science** -- INAOE, Mexico (2006 -- 2009). Thesis: automatic text classification using prototype-based class reduction.
- **Bachelor's in Physics and Mathematics** -- University of Michoacan, Mexico (1998 -- 2003).

Technical Skills

- **Analysis and modeling:** metric definition; evaluation design (baselines, control sets, held-out data, k-fold cross-validation); stratified sampling; recall@k and error-rate metrics; Random Forest, Gradient Boosting, PCA, clustering (HDBSCAN, FINCH); time-series indicators
- **SQL and data:** PostgreSQL (PL/pgSQL, window functions, materialized views, JSONB), SQLite, Oracle, MySQL; ETL, data governance, lineage
- **Languages and libraries:** Python (pandas, NumPy, scikit-learn, SciPy, sentence-transformers), R, SQL
- **AI tooling:** Claude and Claude Code (daily, with runbooks and approval gates), local LLMs (Ollama), retrieval (BM25, embeddings, rank fusion)
- **Reporting:** Apache Superset, Power BI, Tableau, Looker, automated Excel executive reports
- **Certifications:** AWS Cloud Practitioner; AWS Generative AI Fundamentals; Semarchy xDM
- **Languages:** Spanish (native), English (professional), Italian (basic-intermediate), French (basic)